

# MICHAEL PEÑA

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## SUMMARY

Master's-trained statistician with strong experience conducting iterative research projects, performing systems analysis, and applying qualitative and quantitative analytical methods to interpret statistical results. Skilled in computational statistics, database querying and validation, exploratory data analysis, and survey analytics. Experienced in designing, developing, testing, and implementing statistical models; formulating research questions and analysis plans; working to coordinate development of research workflows; and monitoring quality and progress across projects. Able to provide assistance to research teams and provide reports, visualizations, and summaries for stakeholders while maintaining data standards. Brings detail-oriented problem solving and experience presenting findings to diverse audiences. Seeking to contribute subject-matter expertise and data-driven insights in a Research Data Analyst role in Sacramento following upcoming relocation.

## EDUCATION & CERTIFICATES

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|--------------------------------------------------------------------------------------------|---------------|
| California State University, Fullerton   M.S., Statistics                                  | May 2025      |
| California Polytechnic State University, Pomona   B.S., Applied Mathematics and Statistics | May 2023      |
| IBM via edX.org   Certificate of Completion, SQL for Data Science                          | December 2023 |
| IBM via edX.org   Certificate of Completion, Linux Commands & Shell Scripting Basics       | June 2023     |

## SKILLS

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|--------------------|------------------------------------------------------------------------------------------------------------------------------|
| Programming        | R, Python, SQL, Bash, VBA                                                                                                    |
| Databases          | SQL database querying, data management, and validation; software installation and environment configuration on Linux systems |
| Software           | Microsoft Excel, Microsoft 365 Suite, Microsoft Copilot, STATA, R Shiny                                                      |
| Analytical Methods | Statistical Inference, Systems Analysis, Time Series Analysis, Survival Analysis, Cross Validation, Bayesian Data Analysis   |
| Professional       | Technical Writing & Reporting, Data Visualization & Presentation, Project Management, Team Communication, Independent Work   |

## PUBLICATIONS

- Lanza, M.B., Salehpour, S., Addison, O., et al. (2026). A pilot study on aging-related effects on step performance: The role of muscle quality, size, and strength. *Scientific Reports*. <https://doi.org/10.1038/s41598-026-57916-1>  
*Contributing author — conducted the study's statistical analysis and co-authored the analysis and discussion sections.*

## EXPERIENCE

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| <b>Tutor Expert</b>   Mount San Antonio College, Walnut, California                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | November 2023 – Present   |
| <ul style="list-style-type: none"><li>•Provided assistance to 100+ students in manipulating and formatting datasets to remove inconsistencies, and assisted in data interpretation as part of their coursework and projects</li><li>•Supported 100+ students in analyzing quantitative data, interpreting results, and providing feedback for written reports</li><li>•Guided students through statistical analysis methodologies, database concepts, and data visualization techniques using appropriate analytical software</li><li>•Instructed 100+ students in drafting, reviewing, and editing technical reports, maintaining standards for accuracy and clarity so findings were accurately conveyed</li></ul> |                           |
| <b>Statistical Consultant, Dr. Marcel Lanza</b>   University of Maryland School of Medicine (Remote)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | February 2025 – June 2026 |
| <ul style="list-style-type: none"><li>•Coordinated development of the statistical analysis plan with a multidisciplinary research team of 3 graduate researchers and 5 clinical researchers for a pilot study assessing fall risk in older adults, monitoring data quality and project progress through completion</li><li>•Conducted systems analysis and exploratory data analysis of high-dimensional kinematic and ultrasound imaging data, applying statistical diagnostics such as Little's Missing Completely at Random test to assess missingness and inform</li></ul>                                                                                                                                       |                           |

imputation strategy

- Designed, developed, tested, and implemented statistical models including a k-nearest neighbor classification model in powerful statistical software for quantitative data interpretation
- Formulated and modified analysis code and database queries in R and SQL and installed and maintained statistical software environments to support data validation and reproducible research workflows
- Prepared and co-authored the statistical analysis and discussion sections of a peer-reviewed research manuscript, published in Scientific Reports (2026)
- Provided reports and presented findings during stakeholder briefings, delivering data-driven recommendations to guide project direction

**Intern, New Student Programs & Family Engagement** | Cal Poly Pomona, California *July 2023 – September 2023*

- Created clear and detailed itineraries and contact sheets in Microsoft Excel for multiple university orientations, ensuring smooth event coordination with attention to accuracy
- Prepared and printed orientation materials and delivered impactful presentations to large groups of family members and prospective students, educating them about the university and its resources

**Student Assistant & Test Proctor, Disability Resource Center** | Cal Poly Pomona, California *August 2021 – May 2023*

- Applied effective interpersonal communication rules and guidelines with hundreds of students who had learning disabilities
- Created several in-depth operating procedures for future student assistants to follow, maintaining standards for exam administration, documentation, and record keeping

## PERSONAL & ACADEMIC PROJECTS

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**Blood Types Distribution** | Personal Project *October 2025*

- Designed and implemented a complex hierarchical Bayesian sampling model of human blood type distribution across 17 Mexican states, informed by academic research literature

**Electromyogram Prediction** | Academic Project, Time Series Analysis *July 2024*

- Developed, tested, and presented a model to predict the type of active muscle, trained on 2.4+ million observed biostatistical muscle electromyogram datapoints

**Transit Oriented Community Housing Costs** | Academic Project, Statistical Learning *Fall 2024*

- Conducted statistical survival analysis on 300+ datapoints to demonstrate the relationship between delayed construction and affordable housing, modifying and improving methods from previous academic literature

**Building 8, Elevator Wait-Time Prediction** | Academic Project, Statistical Consulting *Spring 2023*

- Designed a statistical model that accurately predicts the wait time of an elevator using on-site quantitative data collection

## RELATED COURSEWORK

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### Statistical Inference and Probability

Fundamentals of probability and mathematical intuition for statistics and statistical analysis

### Statistical Computation

Application of algorithms and simulations using software programming

### Categorical Data Analysis

Training of statistical models to accurately predict qualitative outcomes

### Time Series Analysis

Application of training time-dependent predictive models

### Statistical Learning

Machine learning and predictive training aspects of statistics

### Bayesian Data Analysis

Bayesian statistical methodologies with emphasis on biological and life science applications

## RESEARCH SUMMARY

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As part of my Master's Statistical Consulting Capstone, I worked under clinical researcher Dr. Marcel Lanza on a pilot study examining fall risk in older adults, a project now completed and published in Scientific Reports (2026). I conducted exploratory data analysis and data imputation, built visualizations for trend and risk-factor analysis, applied Principal Component Analysis (PCA) for data reduction, and formulated a K-Nearest Neighbors (KNN) classification model with cross-validation to test and optimize performance. The published 5-nearest neighbor model clarifies the relationships between age, muscle structure, and stepping performance.